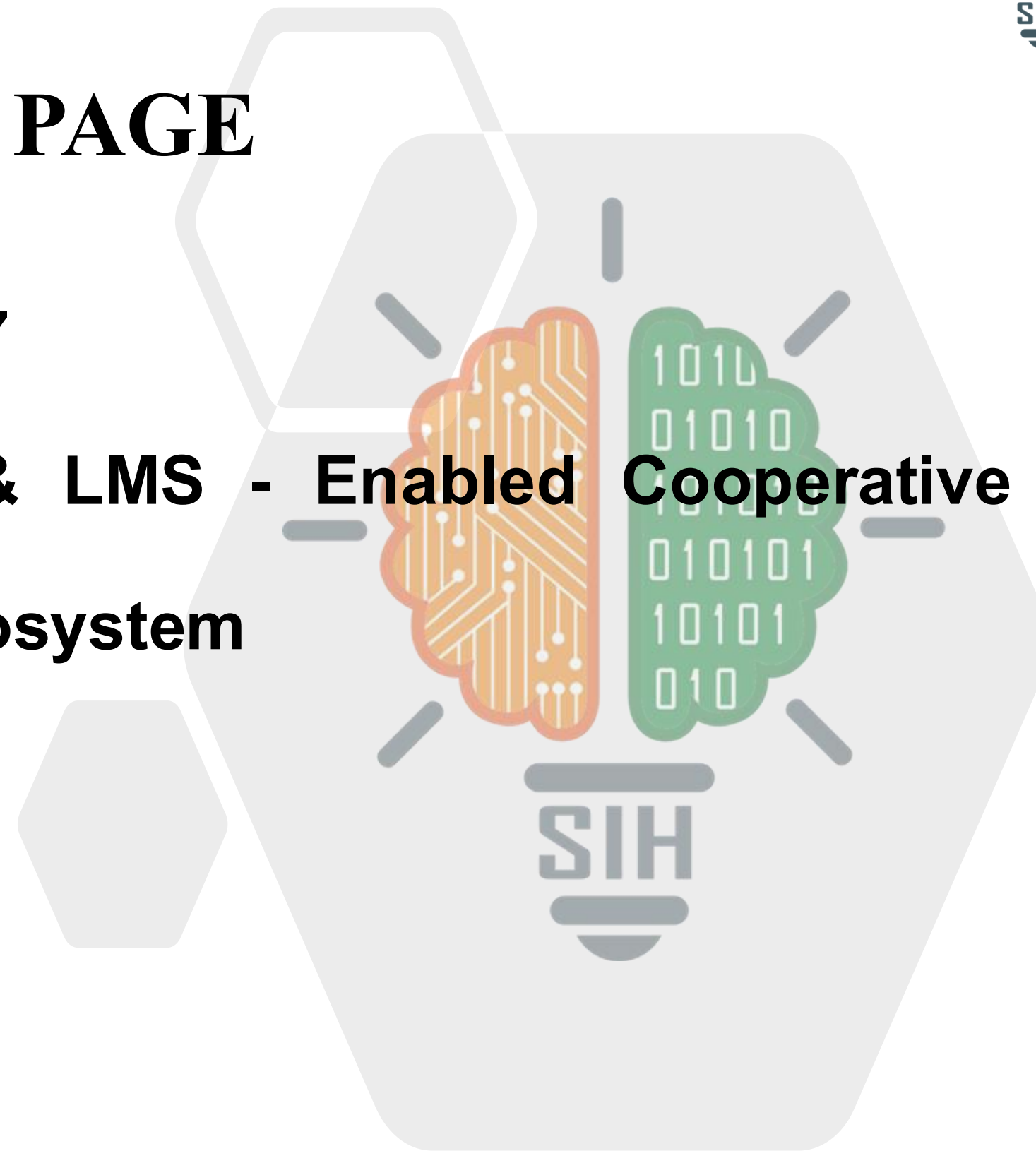


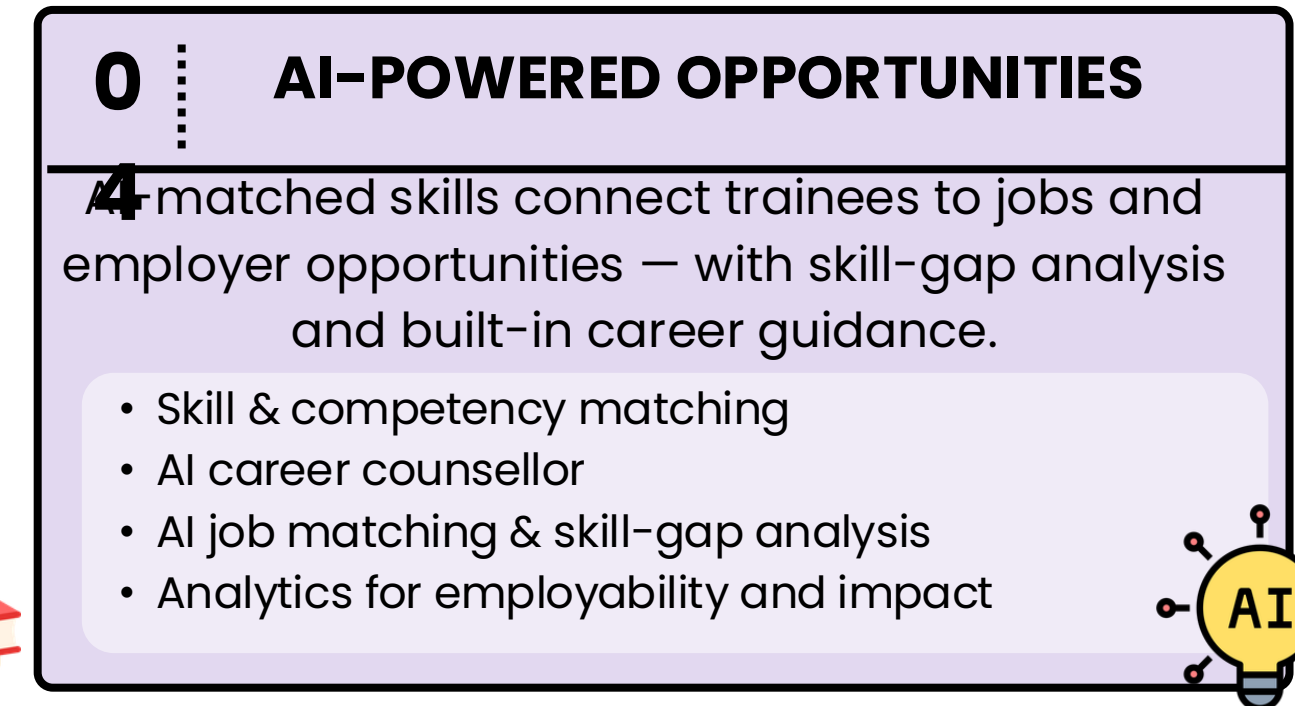
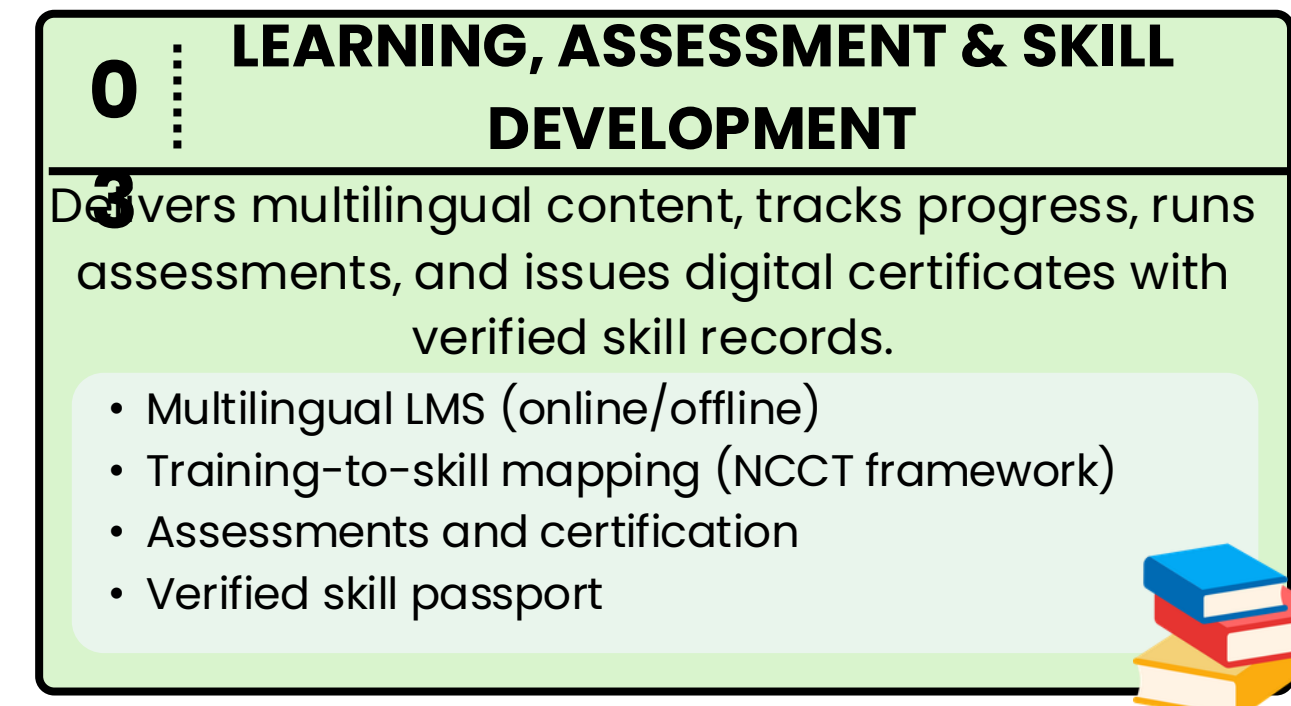
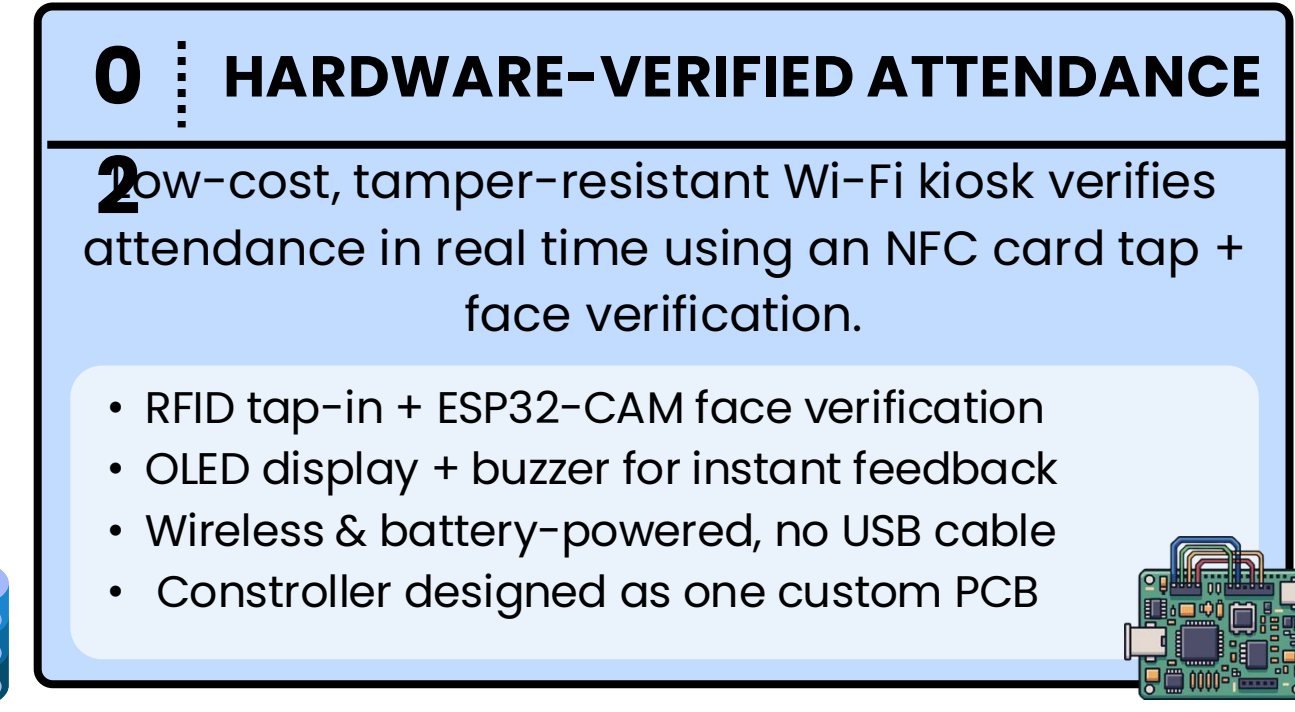
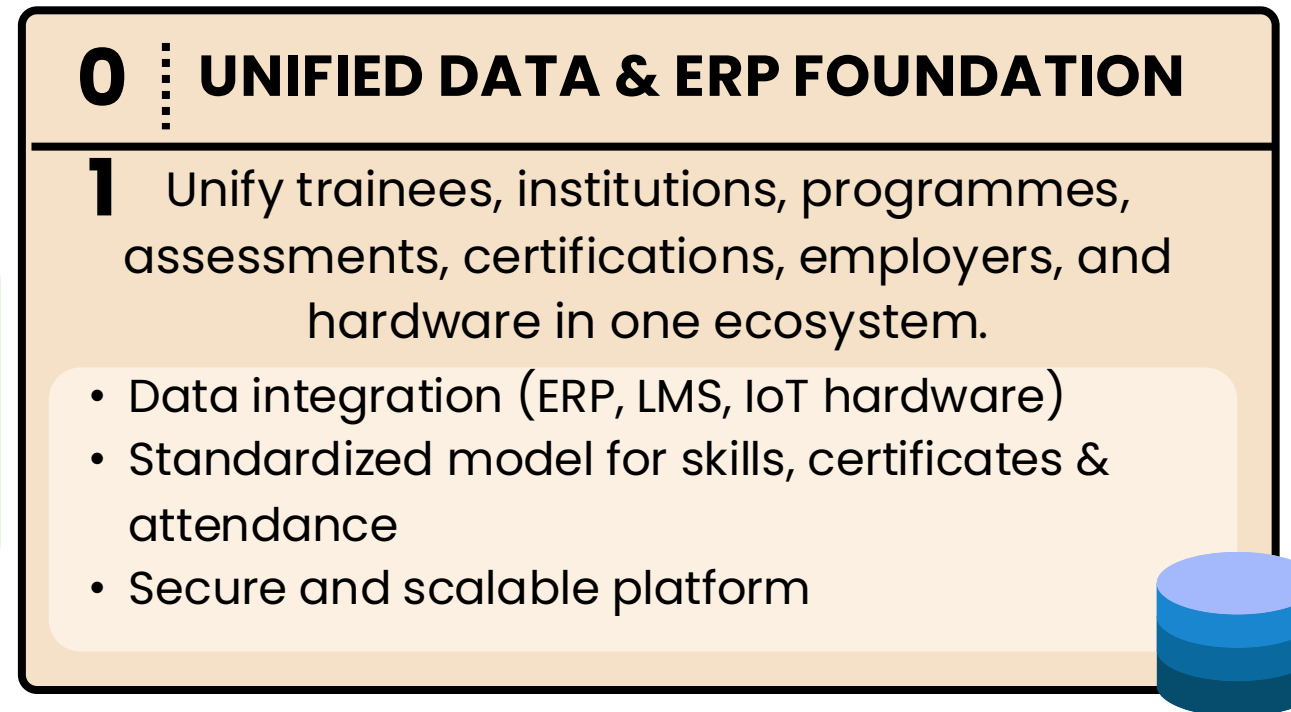
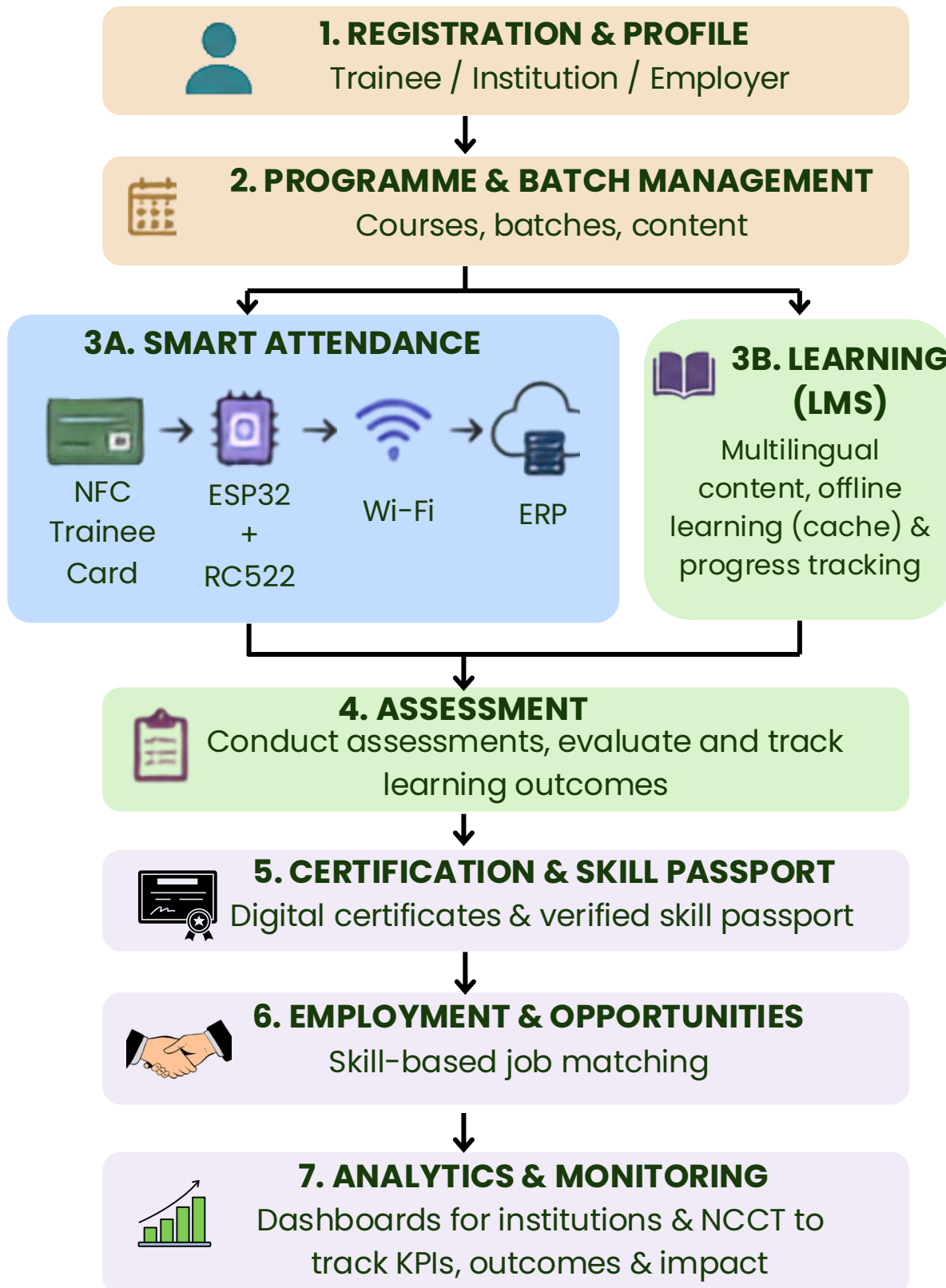
TITLE PAGE

- **Problem Statement ID – SIH26087**
- **Problem Statement Title - AI & LMS - Enabled Cooperative Capacity Building, ERP & Employment Ecosystem**
- **Theme - Smart Education**
- **PS Category - Hardware**
- **Team ID - 130412**
- **Team Name - Untitled**



Proposed Solution

“An integrated platform to train, certify, and connect rural talent with opportunities – powered by AI, LMS, and verified hardware attendance”



FRONTEND

- **Web App:** React 19 + Vite + TypeScript (all 4 role views)
- **Mobile App:** Capacitor build of the same web app (Android + iOS)
- **Kiosk Terminal:** Chrome/Edge page on the kiosk Wi-Fi, driven by the ESP32

BACKEND

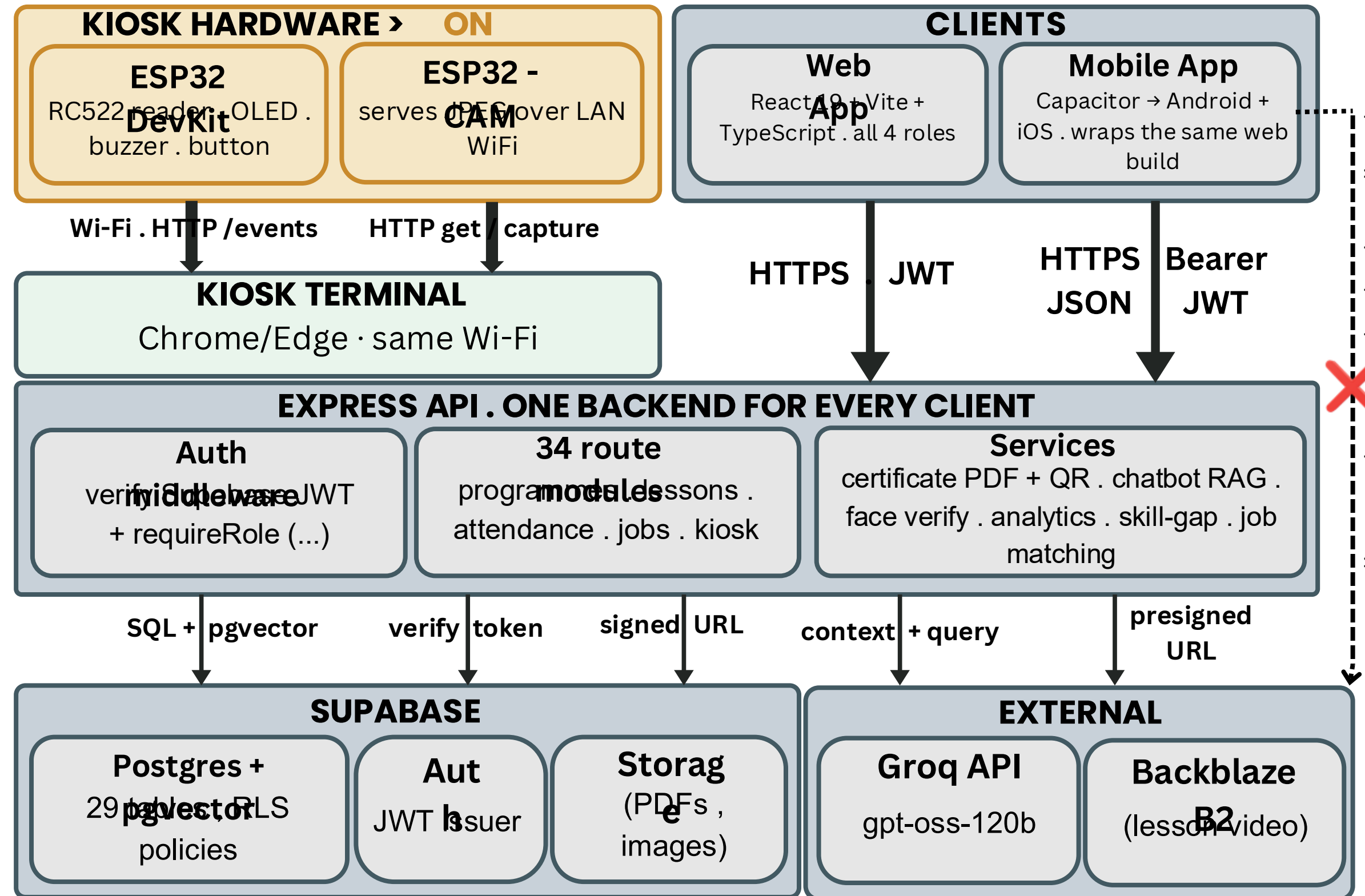
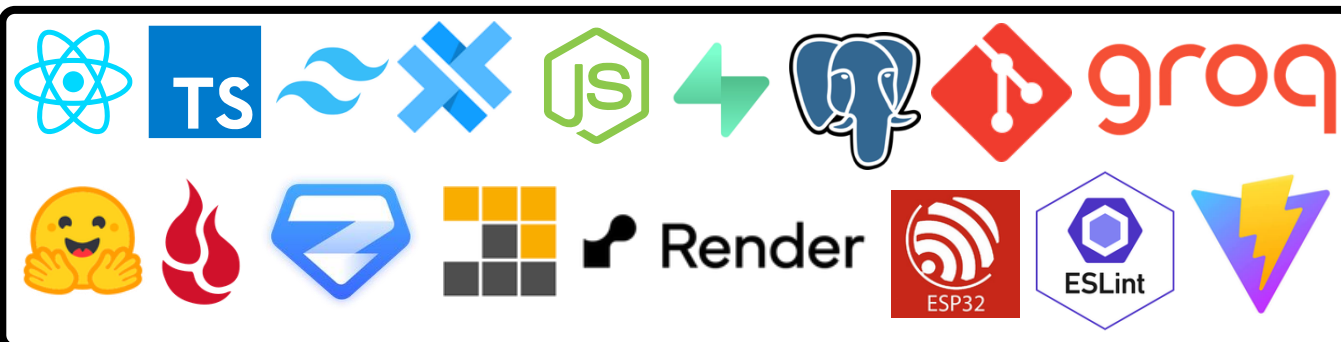
- **Express API:** one backend serving every client
- **Auth middleware:** verifies Supabase JWT + role-based access (requireRole)
- **34 route modules:** programmes, lessons, attendance, jobs, kiosk
- **Core services:** certificate PDF + QR generation, RAG chatbot, face verification, analytics, skill-gap & job matching

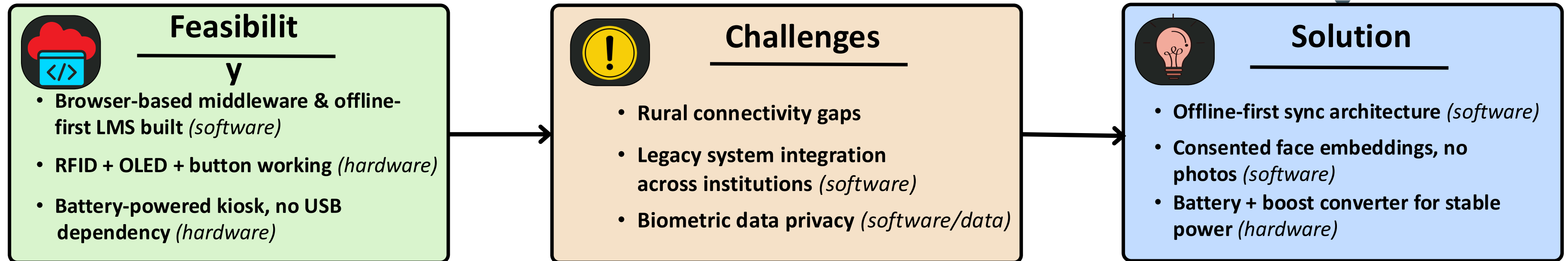
HARDWARE(KIOSK DEVICES)

- **ESP32 DevKit:** RC522 RFID reader, OLED display, buzzer, push button
- **ESP32-CAM** — captures & serves JPEG frames over LAN WiFi for face verification
- **Connects to the Kiosk Terminal** over Wi-Fi (local HTTP), no USB

DATA & INTEGRATIONS

- **Supabase Postgres + pgvector:** 29 tables with row-level security (RLS)
- **Supabase Auth** (JWT issuer) + **Supabase Storage** (PDFs, images)
- **Groq:** RAG chatbot, AI career counsellor & skill-gap
- **Backblaze B2:** stores lesson video content, served via presigned URLs

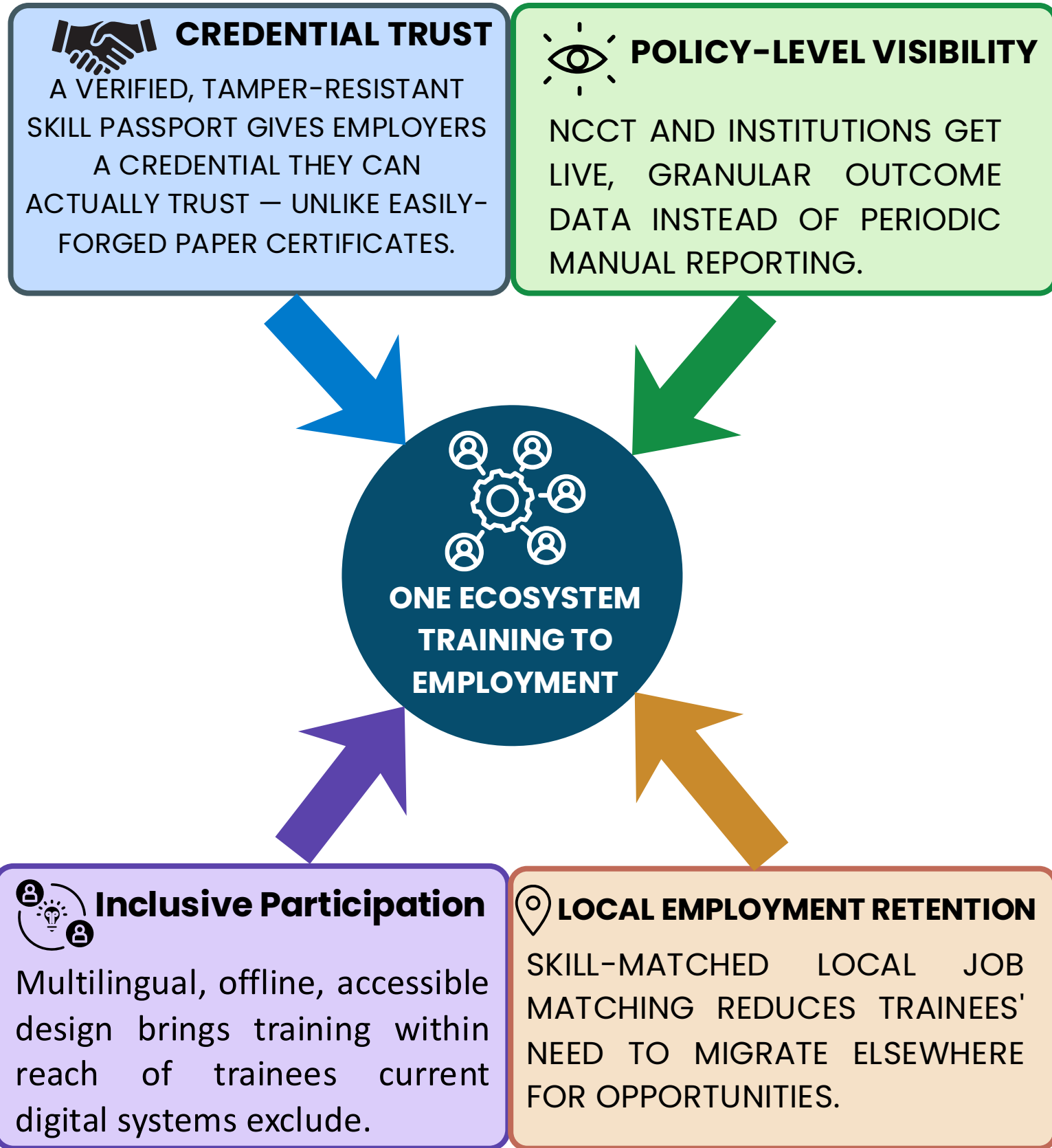




COMPETITIVE EDGE

COST & SCALABILITY		
• ~₹2,000 ESTIMATED HARDWARE COST PER KIOSK • (CUSTOMER) DESIGN CUTS COST FURTHER AT SCALE • (UNIVERSITY) LMS PLATFORM SCALES ACROSS EVERY NCCT INSTITUTION		
TECHNICAL FEASIBILITY	USER EXPERIENCE	
• RC522 RFID: 13.56 MHZ, 0-60MM READ RANGE — CONFIRMED WORKING (HARDWARE) • STANDARD WEB STACK FOR ERP/LMS (SOFTWARE) • ONE PCB CARRIES READER, OLED, BUZZER & BUTTON (HARDWARE)	• CHINESE BACK: ENGLISH-ONLY • OLED + BUZZER, ICON-BASED (HARDWARE) • MULTILINGUAL SUPPORT HANDLED AT THE LMS/WEB LAYER (SOFTWARE) • SIMPLE TAP-AND-SCAN FLOW, MINIMAL TRAINING NEEDED	

Comparison			
Capability	Our Solution	Manual Attendance	Standard Biometric
Prevents proxy attendance	✓	✗	✓
Real-time digital records	✓	✗	✓
Works Offline	✓	✓	✗
Verified Skill Passport	✓	✗	✗
Low Infrastructure Cost	✓	✓	✗



INSTITUTIONAL EFFICIENCY

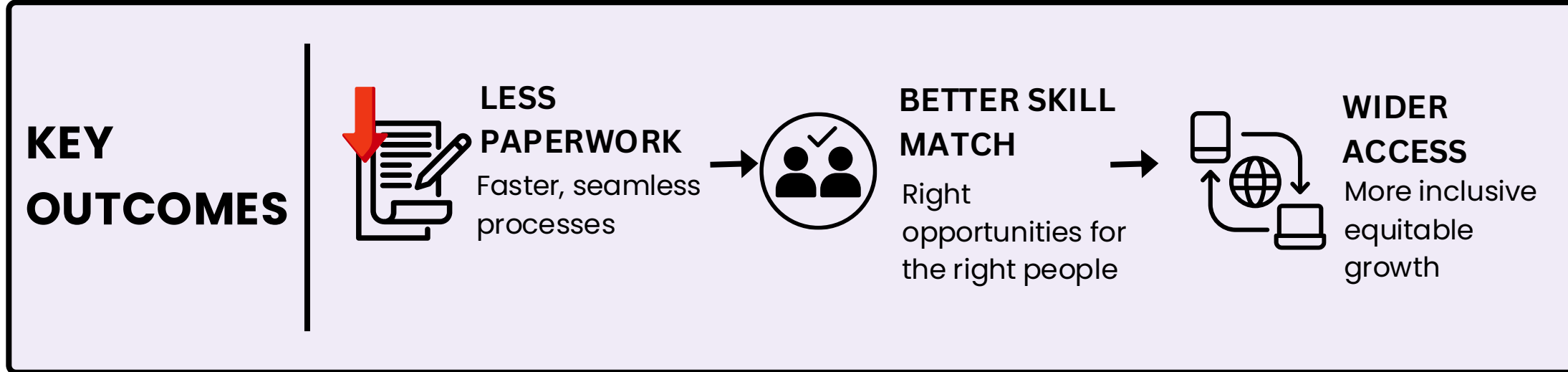
- Unifies registration, batch management, attendance & certification - no more cross checking across separate systems
- Real time institutional dashboards replace manual compilation for tracking KPIs & batch progress

EMPLOYABILITY & LIVELIHOOD

- AI career counsellor gives personalized guidance - most rural trainees have never had access to.
- Skill-matched job recommendations & skill-gap insights connect training directly to outcomes

INCLUSIVE LEARNING ACCESS

- Multilingual LMS removes the language barrier to quality training content
- Offline first design reaches trainees in areas where standard e-learning platforms can't serve
- Accessible interface widens participation beyond who current systems reach



RESEARCH AND REFERENCES



RESEARCH PAPER :-

1) NFC Smart Classroom — Shen et al., 2014

Explores the use of NFC technology for classroom management and student attendance systems.

[NFC Smart Classroom](#)

2) AI Skill-to-Job Recommendation — Giabelli et al., 2021

Investigates AI-driven models that recommend jobs based on individual skills and competencies.

[AI Skill-to-Job Recommendation](#)

3) AI Job Matching Review — Spoladore et al., 2026

A comprehensive review of AI techniques for job matching and employment services.

[AI Job Matching](#)

4) Vocational Training + Job Information in India — Chakravorty et al., 2024

Analyzes the landscape of vocational training and job information systems in India.

[Vocational Training](#)

Prototype link :-

<https://ncct-platform-1.onrender.com>

Demo Video link :-

Documentation link :-

<https://github.com/ShashankPonna/NCCT-platform>

0 INSTITUTIONAL EFFICIENCY

- 1 Unifies registration, batch management, attendance & certification — no more cross-checking across separate legacy systems
- Real-time institutional dashboards replace manual compilation for tracking KPIs & batch progress
- Digital certificate issuance cuts turnaround from weeks of paperwork to instant



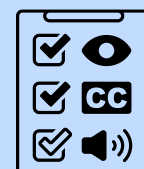
0 EMPLOYABILITY & LIVELIHOOD

- 2 AI career counsellor gives personalized guidance most rural trainees have never had access to
- Verified skill passport builds employer trust — reducing hiring mismatch from unverifiable paper credentials
- Skill-matched job recommendations & skill-gap insights connect training directly to outcomes



0 INCLUSIVE LEARNING ACCESS

- 3 Multilingual LMS removes the language barrier to quality training content
- Offline-first design reaches trainees in areas standard e-learning platforms can't serve
- Accessible interface widens participation beyond who current systems reach



POLICY-LEVEL VISIBILITY

NCCT AND INSTITUTIONS GET LIVE, GRANULAR OUTCOME DATA INSTEAD OF PERIODIC MANUAL REPORTING.



LOCAL EMPLOYMENT RETENTION

SKILL-MATCHED LOCAL JOB MATCHING REDUCES TRAINEES' NEED TO MIGRATE ELSEWHERE FOR OPPORTUNITIES.



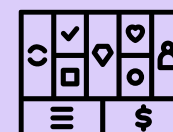
CREDENTIAL TRUST

A VERIFIED, TAMPER-RESISTANT SKILL PASSPORT GIVES EMPLOYERS A CREDENTIAL THEY CAN ACTUALLY TRUST — UNLIKE EASILY-FORGED PAPER CERTIFICATES.



Inclusive Participation

Multilingual, offline, accessible design brings training within reach of trainees current digital systems exclude.



Business Canvas